From Reviews to Insights: Sentiment Analysis and Extractive Summarization of Bali Hotel Reviews

1st Kevin Nathaniel
Computer Science Department, School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
kevin.nathaniel009@binus.ac.id

2nd Rosaline Imely Gosang
Computer Science Department, School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
rosaline.gosang@binus.ac.id

3rd Anthony Davidson Salim
Computer Science Department, School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
anthony.salim@binus.ac.id

4th Joe Kenryoku Fendy
Computer Science Department, School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
joe.fendy@binus.ac.id

***Abstract*—Online hotel reviews provide valuable insights into customer experiences, but their unstructured nature makes large-scale analysis challenging. This study proposes a web-based system for analyzing Bali hotel reviews by integrating sentiment analysis, review summarization, and aspect-based sentiment classification. A transformer-based model, DeBERTaV3, is fine-tuned to classify reviews into positive and negative sentiments, while a pre-trained BART model is used to generate hotel-level abstractive summaries. Aspect extraction is performed using part-of-speech tagging to identify key hotel features, followed by sentiment classification at the aspect level. Experiments conducted on 3,017 TripAdvisor reviews show that the sentiment model achieves an accuracy of 95.20% and an F1-score of 95.94%, indicating strong classification performance. Although ROUGE scores for summarization are relatively low due to limited lexical overlap, BERTScore results demonstrate high semantic similarity between generated summaries and reference texts. The system is deployed using a Streamlit-based interface, highlighting its practical applicability for extracting actionable insights in the hospitality domain.**

Keywords*— *hotel reviews, review summarization, aspect-based sentiment analysis, natural language processing

I. Introduction

Tourism is one of the most dynamic sectors driving global economic growth, with destinations worldwide competing to attract visitors through unique cultural, natural, and recreational experiences [1]. Southeast Asia has seen rapid tourism growth, supported by its diverse landscapes, rich cultures, and strong hospitality industries. Within this region, Bali stands out as one of Indonesia's most iconic destinations, combining traditional heritage, scenic beauty, and modern tourism infrastructure that attract travelers from around the world [2].

Known as the "Island of the Gods," Bali is famous for its tropical scenery, cultural richness, and vibrant hospitality sector [3]. Millions of domestic and international visitors come each year to enjoy its beaches, temples, cuisine, and luxury resorts. According to Indonesia's Central Statistics Agency (BPS), Bali consistently records among the highest tourist arrivals in the country. The number of foreign visitors to Bali has continued to increase between 2019 and 2024, underscoring its lasting global appeal [4]. As tourism remains vital to Bali's economy, the quality of hospitality services, especially hotel accommodations, plays a key role in shaping visitor satisfaction and the island's reputation as a top travel destination.

In today's digital era, travelers increasingly use online platforms to share their opinions and experiences about hotels and other hospitality services [5]. These reviews strongly influence potential visitors' decisions and shape perceptions of hotel quality and brand reputation. Positive reviews attract more bookings, while negative ones highlight areas for improvement [6].

Using sentiment analysis to study these reviews provides a systematic way to uncover patterns and insights from large amounts of text [7]. The results can help hotel managers and tourism stakeholders improve service quality, enhance operations, and offer better guest experiences. Understanding customer sentiment at scale also supports data-driven decisions, personalized marketing, and competitiveness in the hospitality industry [6].

II. Related Works

In recent years, sentiment analysis has become increasingly important in the tourism and hospitality industry, as online reviews serve as a key source of customer feedback and decision-making. Many studies have used natural language processing (NLP) to analyze large amounts of text, helping hotel managers and policymakers better understand guest satisfaction and service quality.

For example, Farisi et al. (2019) used the Multinomial Naïve Bayes algorithm to classify hotel reviews into positive and negative sentiments based on TF–IDF features [8]. While their study showed that Naïve Bayes is a strong baseline for

text classification, it struggled to capture contextual nuances and semantic relationships, particularly when sentiment was implicit or mixed.

To address these limitations, recent studies have turned to deep learning and transformer-based models. Pre-trained models like BERT and its variants have shown strong ability to understand context and capture subtle sentiment [9]. For example, Wen et al. (2023) used BERT and ERNIE to analyze hotel reviews in China, achieving higher accuracy than traditional classifiers. ERNIE, which uses knowledge-enhanced pre-training, performed slightly better, especially in handling complex sentences and implicit sentiments [10]. These findings show that transformer models are effective for sentiment analysis in the hospitality sector, where context and nuanced language matter.

Hence, this study uses DeBERTa (Decoding-enhanced BERT with Disentangled Attention), which improves BERT by separating word content and position embeddings and using an enhanced mask decoder to model word dependencies more effectively [11]. Additionally, by combining extractive summarization, this study not only identifies sentiment polarity but also produces concise summaries of Bali hotel reviews, helping hotel management to enhance their service quality and visitor satisfaction [12].

III. Methodology

To perform sentiment analysis and extractive summarization, several processes need to be carried out. Figure 1 shows the big picture of the end-to-end process of the proposed methodology. The main processes are as follows: (A) Data Preparation and Preprocessing, (B) Model Training, (C) Review Summarization, (D) Aspect Extraction and Sentiment Classification, (E) Evaluation, and (F) Integration and Output. By following these stages, the project aims to achieve accurate and meaningful results for both summarization and sentiment analysis.

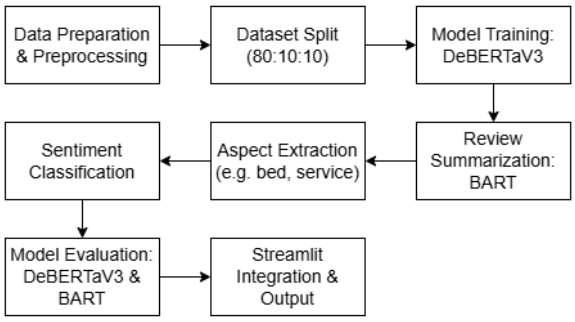

Fig. 1. Overall methodology flow

A. Data Preparation and Preprocessing

The dataset used in this project consists of 3,017 hotel reviews scraped from an online travel platform (TripAdvisor), containing 24 fields such as hotel information, reviewer details, review text, rating, language, and location metadata.

The Review column contains raw, unstructured text that undergoes preprocessing steps, such as tokenization, lowercasing, and noise removal (removing URLs, HTML tags, special characters, extra spaces) to clean and standardize the text before it is used for sentiment analysis. The Rating field will be transformed into a Label column with two sentiment classes: Positive and Negative. A review will be labeled Positive if its rating is above 3, and Negative if its rating is 3 or below.

B. Model Training

In this step, a transformer-based model will be fine-tuned for sentiment classification. DeBERTaV3 is selected for this task as it is one of the top-performing models for sentiment analysis. The dataset will be divided into 80% training, 10% validation and 10% testing sets. This split ensures a balance between model learning and evaluation, leading to more accurate and robust performance. Figure 2 shows the training loop visualization.

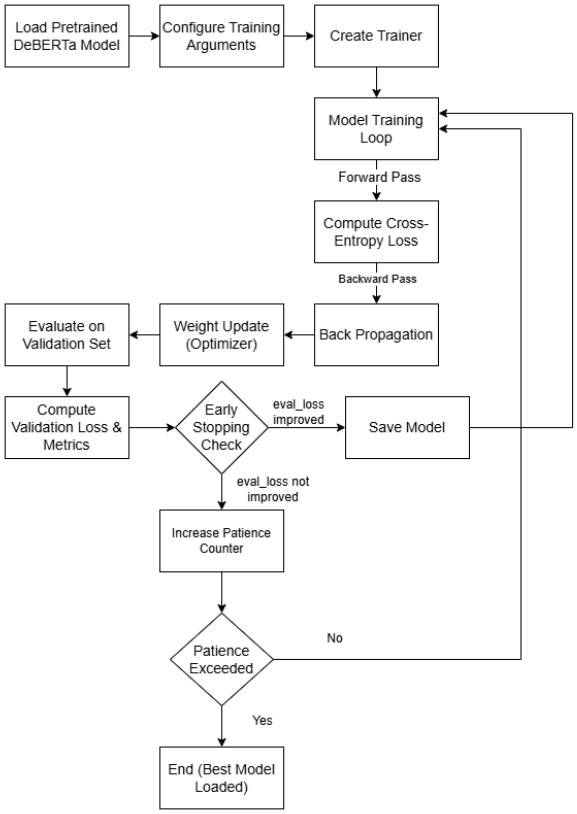

Fig. 2. Training Loop Visualization

C. Review Summarization

In this step, the reviews will be summarized based on the hotel. A pre-trained summarization model called BART (imported from Hugging Face) will be used to summarize the reviews. The results will later be used in the Aspect Extraction and Sentiment Classification step.

D. Aspect Extraction and Sentiment Classification

From the generated summaries, aspect extraction will be performed using the spaCy library. This process utilizes part-of-speech (POS) tagging to identify nouns that represent hotel features, such as bed, food, and service. For instance, given the sentence "The bed is comfortable, but the food tastes bad", the system will extract bed and food as aspects. In addition, the corresponding aspect sentences will be classified using the sentiment classification model trained during the Model Training phase. The final output will present the sentiment polarity of each aspect, such as bed: positive and food: negative.

E. Evaluation

The evaluation process will be conducted in two stages. In the first stage, the performance of the sentiment classification model will be evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. In the second stage, the quality of the generated summaries will be assessed using both ROUGE score and BERTScore.

F. Integration and Output

The user interface will be implemented using Streamlit. It allows users to input a hotel name, after which the system processes the data to produce a summarized review and corresponding sentiment classifications, which are then presented to the user.

IV. Implementation and Results

This section will show the performance of each models used in this study, and the integration of the model.

A. Sentiment Model Evaluation

In order to accurately classify the sentiments of the customer reviews, a transformer model called DeBERTaV3 is used to be trained using the gathered dataset. Table I displays the performance of the model in classifying the customer reviews. Overall, the model shows a relatively good performance with precision as its highest score, The model achieved an accuracy of 95.20%, with precision, recall, and F1-score of 98.61%, 93.42%, and 95.94%, respectively. These results indicate that the model performs well in conducting the sentiment analysis task, especially positive sentiment.

TABLE I
Performance Metrics for DeBERTaV3 Model

Model	Accuracy	Precision	Recall	F1-Score
DeBERTaV3	95.20%	98.61%	93.42%	95.94%

Figure 3 presents the confusion matrix obtained from testing the DeBERTaV3 model. In general, the matrix shows that most classifications are accurate, with majority of the samples correctly classified. Out of 250 data, only 12 are misclassified, with 2 false positive predictions and 10 false negative predictions. Overall, the confusion matrix shows that the model gets most of the predictions correct, indicating a strong classification performance.

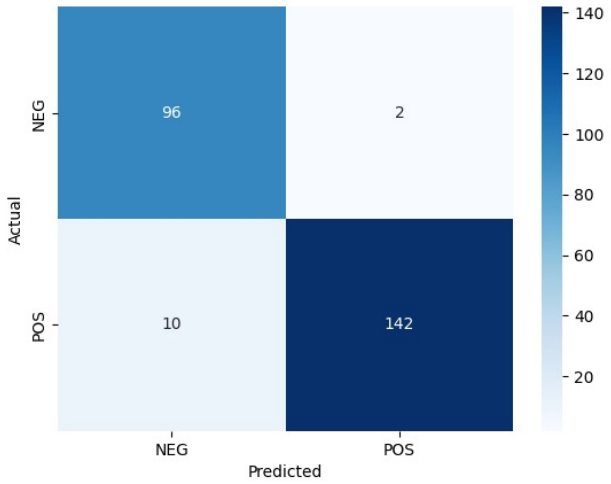

Fig. 3. Confusion matrix of the DeBERTaV3 model

B. Summarization Model Evaluation

Table II displays the ROUGE score and BERTScore of the model used for summarization. The ROUGE-1, ROUGE-2, and ROUGE-L scores are relatively low, while the BERTScore suggests that the model shows a high semantic similarity between the generated summary and reference text.

TABLE II
Summarization Evaluation Metrics

Metric	Score
ROUGE-1	0.1099
ROUGE-2	0.0180
ROUGE-L	0.0712
BERTScore (F1)	0.8202

C. Model Integration

The system's user interface is deployed on a website created using Streamlit, which serves as its user interface (website link: https://balihotel-review-analyzer.streamlit.app/). On the website, users are able to select which hotel review they want to summarize. Users are not required to insert reviews, as the system itself already have the joined reviews. Other than that, users have three options for the length of the summary, which are short, medium, and long. Figure 4 shows the website when a user is selecting the hotel and length of summary.

When users click on the "Summarize Review" button, the website will output the generated summary and the aspect sentiment analysis. Figure 5 shows the output of the website. As displayed in the figure, the generated summary is relatively short in length. In addition to the general summary, it also outputs the sentiment and summary of each aspect found when summarizing the reviews.

V. Discussion and Limitation

Table I shows that the model shows a relatively good performance with precision as its highest score, which indicates that the model is effective at identifying positive sentiment. Even

though the model have recall as its lowest score, it remains robust and can still accurately classify the inputs.

Analysis of the confusion matrix suggests that these errors happened due to the context of the reviews, which might include sarcasm. Sarcasm expressions can lead to misidentification because it tricks the model into thinking that the review have positive sentiment, while the customer intends to give negative review.

Although the sentiment model achieved high performance, the effectiveness of the program also depends on the summaries generated by the BART model. To assess this, ROUGE scores are used to measure the quality of summaries. ROUGE score works by comparing the generated summary with another summary, and in this case extractive summaries are used. Overall, the ROUGE scores of the model are relatively low, indicating minimal lexical overlap between the generated and the referenced summary. While ROUGE-1 suggests that the model captures some key terms, the very low ROUGE-2 score indicate limited phrase-level similarity. Likewise, the ROUGE-L shows minimal structural alignment with the referenced summaries. These phenomenons are likely happened due to difference of how the summaries are generated, where BART tends to generate paraphrased and abstract summaries, and extractive summaries copy exact phrases from source. The difference of both summaries contribute most to why the ROUGE scores are low because ROUGE score evaluates the similarity of the summarized sentence, not the quality.

In order to address this problem, BERTScore was utilized to measure the sematic similarity between the summaries. Unlike ROUGE score, BERTScore compute the conceptual similarity between the texts, which is more suitable for evaluating paraphrased texts. The BERTScore shown in table II indicate that the generated summaries preserved the meaning of the original reviews despite using different words. This shows that the BART model successfully captured the information from the user reviews.

VI. Conclusion and Future Work

This study proposes a web-based system which analyzes hotel reviews in Bali by combining sentiment analysis, extractive summarization, and aspect-based sentiment analysis. DeBERTaV3 is utilized to perform the sentiment analysis tasks, and it demonstrated a strong performance by achieving high evaluation metric score. This indicates that the model is effective in classifying the sentiment expressed in the reviews.

In addition to sentiment analysis, review summarization is also performed in this study. A pre-trained model call BART is used to generate a summary based on the hotel reviews. Although the Rouge scores are relatively low, the model showed great performance when evaluated using BERTScore. This indicates that the model is successful in preserving the core meaning of the reviews while also paraphrasing the text. Furthermore, aspect-based sentiment analysis allows the system to provide a more detailed description on each aspect that is highlighted in the reviews.

Although the study shows promising results, there are still limitations that cease to exists in this study. The DeBERTaV3 model that is used to conduct sentiment analysis struggled with classifying sarcastic reviews, and the summarization quality still depend on the consistency of the reviews for each hotel. Reflecting on these limitations, future works should focus on utilizing models that are aware of sarcasm, and extending the system to support multilingual reviews. Overall, this study shows that combining sentiment analysis, generated summarization, and aspect-based sentiment analysis, can effectively help users on gaining insights from online hotel reviews.

References

[1] H. B. Gidebo, "Factors determining international tourist flow to tourism destinations: A systematic review," *Journal of Hospitality Management and Tourism*, vol. 12, no. 1, pp. 9–17, 2021.

[2] S. Priatmoko, M. Kabil, R. Magda, E. Pallas, and L. D. David, "Bali and the next proposed tourism development model in indonesia," *Regional Science Inquiry*, vol. 13, no. 2, pp. 161–180, 2021.

[3] M. I. Kersapati, "Spatio-historical data enrichment for toponomastics in bali, the island of gods," *GeoJournal*, vol. 88, no. 5, pp. 5489–5510, 2023.

[4] Badan Pusat Statistik – Provinsi Bali, "Number of foreign visitors arriving directly by nationality to bali 2019–2024," https://bali.bps.go.id/en/statistics-table/1/MTkzIzE=/number-of-foreign-visitors-arriving-directly-by-nationality-to-bali-2019-2024.html, 2025, accessed: 2025-10-17.

[5] K. Dutta, K. Sharma, and T. Goyal, "Customer's digital advocacy: the impact of reviews and influencers in building trust for tourism and hospitality services," *Worldwide Hospitality and Tourism Themes*, vol. 13, no. 2, pp. 260–274, 2021.

[6] Y.-F. Chen, R. Law, and K. K. Yan, "Negative ewom management: how do hotels turn challenges into opportunities?" *Journal of Quality Assurance in Hospitality & Tourism*, vol. 23, no. 3, pp. 692–715, 2022.

[7] J. Cui, Z. Wang, S.-B. Ho, and E. Cambria, "Survey on sentiment analysis: evolution of research methods and topics," *Artificial Intelligence Review*, vol. 56, no. 8, pp. 8469–8510, 2023.

[8] A. A. Farisi, Y. Sibaroni, and S. Al Faraby, "Sentiment analysis on hotel reviews using multinomial naïve bayes classifier," in *Journal of Physics: Conference Series*, vol. 1192, no. 1. IOP Publishing, 2019, p. 012024.

[9] F. A. Acheampong, H. Nunoo-Mensah, and W. Chen, "Transformer models for text-based emotion detection: a review of bert-based approaches," *Artificial Intelligence Review*, vol. 54, no. 8, pp. 5789–5829, 2021.

[10] Y. Wen, Y. Liang, and X. Zhu, "Sentiment analysis of hotel online reviews using the bert model and ernie model—data from china," *PLOS ONE*, vol. 18, no. 3, pp. 1–14, 03 2023. [Online]. Available: https://doi.org/10.1371/journal.pone.0275382

[11] P. He, X. Liu, J. Gao, and W. Chen, "Deberta: Decoding-enhanced bert with disentangled attention," *arXiv preprint arXiv:2006.03654*, 2020.

[12] Y. Liu, "Fine-tune bert for extractive summarization," *arXiv preprint arXiv:1903.10318*, 2019.

Appendix

A. Team Contribution

1) **Anthony Davidson Salim**
   - Led the design and implementation of the DeBERTaV3 and BART model.
   - Assisted in dataset preprocessing and hyperparameter tuning.
   - Participated in system testing and performance evaluation.
   - Contributed to writing the methodology and dataset description sections.

2) **Kevin Nathaniel**

- Contributed to dataset preparation, including data scraping.
- Assisted in data preprocessing and model testing.
- Contributed to the implementation and results sections of the report.
- Participated in model training and evaluation.

3) **Rosaline Imely Gosang**
   - Developed and integrated the web-based user interface for the review summarizer and aspect sentiment analyzer.
   - Assisted in dataset preparation.
   - Participated in report review and system validation.
   - Managed documentation structure and report formatting.
4) **Joe Kenryoku Fendy**
   - Conducted performance evaluation and metric analysis.
   - Assisted in data preprocessing and model testing.
   - Managed documentation structure and report formatting.
   - Contributed to the discussion, limitations, and conclusion sections.

B. Streamlit Screenshots

Fig. 4. User Interface for the Deployed Model

C. Code Snippets

Fig. 5. User Interface for the Deployed Model

```
# Load the model
num_labels = 2    # POSITIVE & NEGATIVE

model = AutoModelForSequenceClassification.from_pretrained(
    "microsoft/deberta-v3-base",
    num_labels = num_labels
).to(DEVICE)
```

Fig. 6. Load Sentiment Model

```
# Define Training Arguments
training_args = TrainingArguments(
    output_dir="deberta_model",
    eval_strategy="epoch",
    save_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    num_train_epochs=10,    # can be high because early stopping will automatically stop it.
    weight_decay=0.01,
    logging_steps=50,
    load_best_model_at_end=True,
    metric_for_best_model="eval_loss",
    greater_is_better=False,
    push_to_hub=False,
    seed=42,
    report_to = "none",

    # To make training faster & lower GPU memory usage
    fp16=True,              # <-- enable mixed precision training
    fp16_full_eval=True     # <-- use FP16 during evaluation too

)
```

Fig. 7. Training Arguments

```
# Create Trainer
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized_dataset["train"],
    eval_dataset=tokenized_dataset["validation"],
    tokenizer=tokenizer,
    compute_metrics=compute_metrics,
    callbacks=[EarlyStoppingCallback(early_stopping_patience=3)]
)
```

Fig. 8. Trainer

```python
# Predict the test set

# Get raw predictions
predictions = trainer.predict(tokenized_dataset["test"])
print(predictions)

# Convert logits to class labels (0 or 1)
y_pred = np.argmax(predictions.predictions, axis=1)

# True labels
y_true = predictions.label_ids
```

Fig. 9. Sentiment Model Evaluation

```python
def simple_extractive_summary(text, num_sentences=2):
  sentences = sent_tokenize(text)

  # Create word frequency
  words = word_tokenize(text.lower())
  words = [word for word in words if word not in stop_words and word.isalnum()]
  freq = Counter(words)

  # Score each sentence
  sentence_scores = {}
  for sent in sentences:
    for word in word_tokenize(sent.lower()):
      if word in freq:
        sentence_scores[sent] = sentence_scores.get(sent, 0) + freq[word]

  # Pick top N sentences
  top_sentences = sorted(sentence_scores, key=sentence_scores.get, reverse=True)[:num_sentences]

  return " ".join(top_sentences)
```

Fig. 10. Simple Extractive Summarization Function

```python
# Load summarization model
summarizer = pipeline(
    "summarization",
    model="facebook/bart-large-cnn",
    tokenizer="facebook/bart-large-cnn",
    device = 0 if torch.cuda.is_available() else -1
)
```

Fig. 11. Load Summarization Model

```python
def safe_bart_summarize(text):
    try:
        # Hard limit input size
        text = text[:1000]

        summary = summarizer(
            text,
            max_length=60,
            min_length=20,
            do_sample=False
        )
        return summary[0]["summary_text"]

    except Exception as e:
        return "Summary failed"
```

Fig. 12. BART Summarization

```python
# ROUGE score
scorer = rouge_scorer.RougeScorer(['rouge1', 'rouge2', 'rougeL'], use_stemmer=True)

rouge1_scores = []
rouge2_scores = []
rougeL_scores = []

for simple_extractive_sum, bart_sum in zip(grouped_reviews['summary'], grouped_reviews['bart_summary']):
  score = scorer.score(simple_extractive_sum, bart_sum)

  rouge1_scores.append(score['rouge1'].fmeasure)
  rouge2_scores.append(score['rouge2'].fmeasure)
  rougeL_scores.append(score['rougeL'].fmeasure)

print("ROUGE-1:", np.mean(rouge1_scores))
print("ROUGE-2:", np.mean(rouge2_scores))
print("ROUGE-L:", np.mean(rougeL_scores))
```

Fig. 13. ROUGE scores

```python
# BERTScore (Semantic Similarity)
P, R, F1 = bert_score_func(
    grouped_reviews["bart_summary"].tolist(),
    grouped_reviews["summary"].tolist(),
    lang="en",
    device = "cuda" if torch.cuda.is_available() else "cpu"
)
```

Fig. 14. BERTScore Evaluation

```python
def aspect_sentiment_pipeline(summary_text):
    aspects = extract_aspects(summary_text)
    aspect_sentences = get_aspect_sentences(summary_text, aspects)

    results = {}

    for asp, sentences in aspect_sentences.items():
        # classify each sentence about this aspect
        sentiments = [predict_sentiment(s) for s in sentences]

        # majority vote (if multiple sentences mention the same aspect)
        final_sentiment = max(set(sentiments), key=sentiments.count)

        results[asp] = final_sentiment

    return results
```

Fig. 15. Aspect Extraction Sentiment Classification Function